

WhatsApp Agent — Complete Project Documentation

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1. Project Overview

WhatsApp Agent is an internal employee task management system for ~500 staff, powered by WhatsApp messaging via the Twilio API. Admins assign tasks via @mentions, employees confirm completion or request help, and the system handles follow-ups, escalation, and knowledge base lookups — all through WhatsApp.

Tech Stack

Layer	Technology
Backend	FastAPI (Python 3.12+)
Database	PostgreSQL 16 + pgvector
Cache/Queue	Redis

Frontend	React 18 + Vite + Tailwind CSS
Messaging	Twilio WhatsApp API
NLP	Anthropic Claude / OpenAI (keyword fallback without API key)
Deployment	Docker Compose

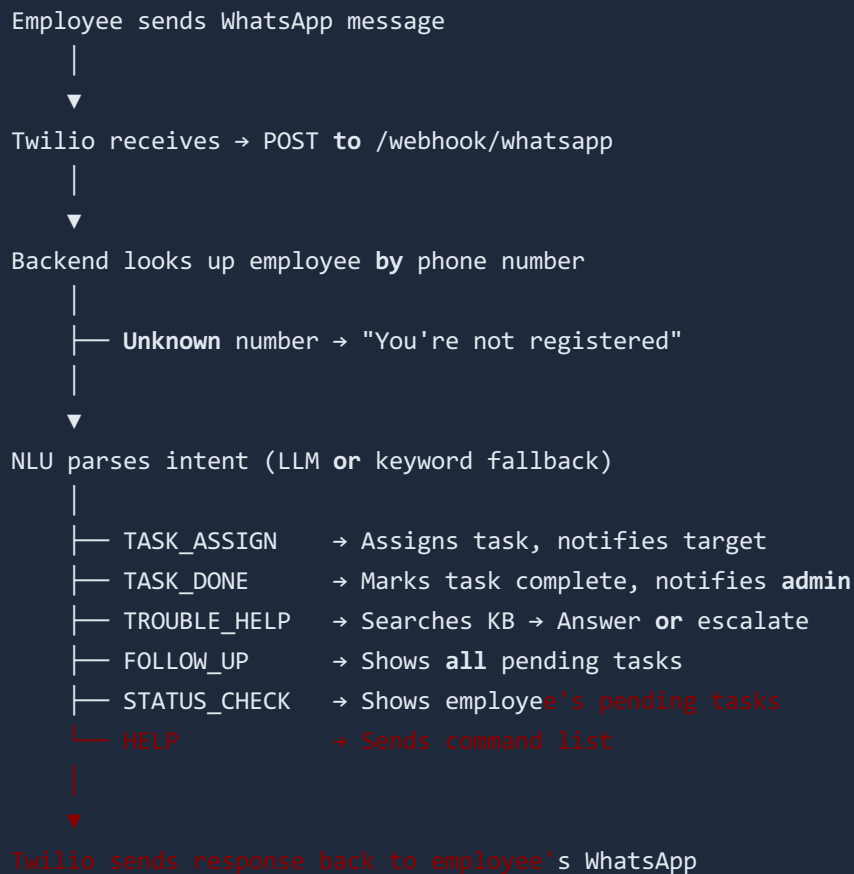
Key Capabilities

- **Task Assignment:** Admins assign tasks via WhatsApp @mentions
- **Multi-language Support:** English, Hindi, Hinglish, Gujarati detection
- **Auto Follow-ups:** Configurable periodic task reminders
- **RAG Knowledge Base:** Employees get instant answers from company docs
- **Escalation Workflows:** Unresolved issues escalate to admins
- **Admin Dashboard:** Full React SPA for monitoring and management

2. Architecture



Message Flow



3. Features

3.1 Task Management

- **Assign tasks** via @mention in WhatsApp messages
- **Automatic priority detection** (high/medium/low from keywords)
- **Due date parsing** from natural language ("by Friday", "by tomorrow")
- **Mark tasks complete** by replying "done", "ho gaya", "kar diya"
- **View pending tasks** via "my tasks", "all tasks", "team pending"

3.2 Multi-language NLP

Language	Detection Method
English	Default fallback
Hindi	Hinglish words (hai, nahi, kya, karna, etc.)
Hinglish	Hindi words + Latin script characters

3.3 Knowledge Base (RAG)

- **Upload PDF/TXT documents** via admin dashboard
- **Vector embeddings** stored in pgvector
- **Semantic search** against employee questions
- **Past resolutions** stored for future reference
- **Fallback to keyword matching** when no LLM API key

3.4 Escalation & Follow-ups

- **Auto-escalation** when KB can't answer a help request
- **Admin notifications** for all escalations
- **Configurable follow-up intervals** (periodic reminders)
- **Overdue task detection** with automated reminders

3.5 Admin Dashboard

- **Real-time metrics** (employees, pending tasks, completed)
- **Employee management** (CSV import, search, department filter)
- **Task monitoring** (status filter, priority filter, auto-refresh)
- **Knowledge base management** (upload, search)
- **Twilio status monitoring** (dev vs production mode)
- **JWT authentication** with token persistence

4. Installation & Setup

4.1 Prerequisites

- Python 3.12+ (3.14 supported but may need C-extension wheels)
- Node.js 20+ and npm
- Docker + Docker Compose (for production or PG with pgvector)
- PostgreSQL 16 with pgvector extension (Docker recommended)

4.2 Backend Setup

```
cd wa-bot/backend

# Create virtual environment
python -m venv venv
.\venv\Scripts\activate      # Windows
source venv/bin/activate     # Linux/Mac

# Install dependencies
pip install -r requirements.txt

# Configure environment
cp ../../.env.example .env
# Edit .env with your values

# Start server
uvicorn app.main:app --host 0.0.0.0 --port 8000 --reload
```

4.3 Frontend Setup

```
cd wa-bot/frontend

# Install dependencies
npm install

# Start dev server
npm run dev
```

Open <http://localhost:5173> → Login with `admin` / `admin123`

4.4 Database Setup

Option A: Existing Docker container (dev)

```
# Use existing hrms-db container on port 5433
# Ensure pgvector extension is installed:
docker exec hrms-db psql -U wabot -d wabot -c "CREATE EXTENSION IF NOT EXISTS vector;"
```

Option B: Docker Compose (production)

```
docker-compose up -d db
# Creates fresh PostgreSQL 16 + pgvector on port 5432
```

4.5 Import Sample Employees

```
curl -X POST http://localhost:8000/api/employees/import \
-F "file=@sample_employees.csv"
```

The CSV file has these 8 sample employees:

Name	Department	Role	WhatsApp	Admin
Raj Patel	Engineering	Developer	+919876543210	No
Priya Sharma	Engineering	Lead	+919876543211	Yes
Amit Singh	Support	Engineer	+919876543212	No
Sneha Desai	HR	Manager	+919876543213	Yes
Vikram Mehta	Finance	Analyst	+919876543214	No
Kiran Joshi	Engineering	Senior Developer	+919876543215	No
Pooja Rao	Marketing	Coordinator	+919876543216	No
Arjun Nair	Operations	Supervisor	+919876543217	No

5. Twilio Configuration

5.1 Get Credentials

1. **Sign up:** console.twilio.com (<https://console.twilio.com/>)
2. **Account SID:** Dashboard → Account Info → Account SID
3. **Auth Token:** Dashboard → Account Info → Auth Token → View
4. **WhatsApp Sandbox Number:** Messaging → Try it out → Send a WhatsApp message

5.2 Join Sandbox (Testing Only)

From your WhatsApp, send to the sandbox number:

```
join <your-sandbox-word>
```

Example: `join two-banana`

5.3 Configure Environment

```
cp .env.example .env
```

Edit `.env` :

```
TWILIO_ACCOUNT_SID=ACxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxx
TWILIO_AUTH_TOKEN=your_auth_token_here
TWILIO_WHATSAPP_NUMBER=+14155238886
```

5.4 Set Webhook URL

Local testing (ngrok):

```
ngrok http 8000
# Output: https://abc123.ngrok.io → http://localhost:8000
```

In Twilio Console → WhatsApp → Sandbox Settings:

- **"When a message comes in":** `https://abc123.ngrok.io/webhook/whatsapp`

Production:

- Set webhook to `https://yourdomain.com/webhook/whatsapp`

5.5 Verify Configuration

```
curl http://localhost:8000/api/twilio/status
# {"configured": true, "mode": "production", ...}
```

5.6 Send Test Message

```
curl -X POST http://localhost:8000/api/twilio/test \
-H "Content-Type: application/json" \
-d '{"to_number": "+919876543210", "message": "Test from WA Bot!"}'
```

6. WhatsApp Chat Examples

6.1 Admin Assigns Task

 Priya Sharma (Admin) → Bot:
@Raj fix the login bug high priority

 Bot → Raj Patel (+919876543210):
 *New Task Assigned*

From: Priya Sharma
Task: @Raj fix the login bug high priority
Priority: HIGH

Reply 'done' when complete, or describe your issue for help.

 Bot → Priya Sharma (+919876543211):
 Task assigned to Raj Patel: "@Raj fix the login bug high priority"

6.2 Admin Assigns Task with Due Date

 Sneha Desai (Admin) → Bot:
@Amit configure the new router by Friday high

 Bot → Amit Singh:
 *New Task Assigned*

From: Sneha Desai
Task: @Amit configure the new router by Friday high
Priority: HIGH
Due: 19 May 2026

Reply 'done' when complete, or describe your issue for help.

6.3 Employee Marks Task Complete

 Raj Patel → Bot:
done

 Bot → Raj Patel:
 '@Raj fix the login bug high priority' marked as done! Great work 🎉

 Bot → Priya Sharma:
 Raj Patel completed: "@Raj fix the login bug high priority"

6.4 Employee Checks Own Tasks

 Raj Patel → Bot:
my tasks

 Bot → Raj Patel:

 *Your pending tasks:*

1. Fix database migration [high] (Due: 20 May)
2. Review PR #142 [medium]
3. Update API docs [low] (Due: 25 May)

6.5 Admin Checks Team Tasks

 Priya Sharma (Admin) → Bot:
all tasks

 Bot → Priya Sharma:

 *All Pending Tasks:*

- Fix database migration → Raj Patel [high]
- Review PR #142 → Raj Patel [medium]
- Configure new router → Amit Singh [high]
- Prepare Q2 report → Vikram Mehta [medium]
- Social media campaign → Pooja Rao [low]

6.6 Employee Asks for Help (KB Search)

 Amit Singh → Bot:
How do I reset the printer network settings?

 Bot → Amit Singh:

 *Solution found:*

To reset printer network settings:

1. Go to Menu → Network → Reset
2. Select "Factory Default Network Settings"
3. Confirm and wait for restart
4. Reconfigure WiFi through the printer panel

If this didn't help, reply 'escalate' for admin assistance.

6.7 Employee Asks for Help (Escalation)

 Amit Singh → Bot:
The server keeps crashing after deployment

 Bot → Amit Singh:
✗ I couldn't find a solution. I'm notifying the admin.
Someone will help you shortly.

 Bot → Priya Sharma (Admin):
 *Escalation:* Amit Singh needs help!
Query: The server keeps crashing after deployment
Ticket #: a1b2c3d4

 Bot → Sneha Desai (Admin):
 *Escalation:* Amit Singh needs help!
Query: The server keeps crashing after deployment
Ticket #: a1b2c3d4

6.8 Admin Requests Help Command List

 Priya Sharma (Admin) → Bot:
help

 Bot → Priya Sharma:
 *WhatsApp Agent – Admin Commands*

 *Assign task:* @name task description due date priority
Example: @Raj fix server bug by Friday high priority

 *Employee replies done/ho gaya* → auto-marks complete

 *Help/Stuck* → bot searches KB, escalates to you

 *My team tasks:* 'all tasks', 'team pending', 'sabka status'

 *Follow up:* 'follow up', 'check all pending'

 *Escalations notified to you automatically*

6.9 Employee Requests Help Command List

 Raj Patel → Bot:

help

 Bot → Raj Patel:

 *WhatsApp Agent – Employee Commands*

 *Done:* 'done', 'ho gaya', 'complete', 'kar diya'

 *Help:* 'stuck', 'error', 'samajh nahi aaya', 'help'

 *My tasks:* 'my tasks', 'pending', 'kya karna hai'

Reply to task messages with any issue and bot will try to help!

6.10 Multi-language Support

 Raj Patel → Bot (Hinglish):

bhai mera kya pending kaam hai?

 Bot → Raj Patel (Hinglish detected):

 *Your pending tasks:*

1. Fix database migration [high] (Due: 20 May)

2. Review PR #142 [medium]

 Raj Patel → Bot (Hinglish completion):

ho gaya sab

 Bot → Raj Patel:

 'Fix database migration' marked as done! Great work 🎉

6.11 Unregistered Number

 +919999999999 → Bot:

hello

 Bot → +919999999999:

✗ You're not registered in our system.

Please contact your admin to get registered.

7. Admin Dashboard Guide

7.1 Login

- Navigate to `http://localhost:5173`
- Default credentials: `admin / admin123`
- Session persists for 24 hours via JWT in `localStorage`

7.2 Dashboard Page

- **Employee count** — Total registered employees
- **Pending tasks** — Tasks awaiting completion
- **Completed tasks** — Tasks marked done
- **WhatsApp mode** — Shows DEV or PRODUCTION
- **System status** — Backend, PostgreSQL, Redis connection status
- **Quick Start guide** — Onboarding steps

7.3 Employees Page

- **Import CSV** — Upload employee CSV file
- **Search** — Filter by name, role, or WhatsApp number
- **Department filter** — Filter by department dropdown
- **Table** — Shows Name, Department, Role, WhatsApp, Admin status

7.4 Tasks Page

- **Search** — Filter tasks by title text
- **Status filter** — All / Pending / In Progress / Done
- **Priority filter** — All / High / Medium / Low
- **Auto-refresh** — Updates every 30 seconds
- **Task count** — Shows filtered vs total count

7.5 Knowledge Base Page

- **Upload documents** — PDF or TXT files with title
- **Search** — Semantic search against uploaded documents
- **Results** — Shows title, content snippet, similarity score

8. API Reference

Authentication

Method	Path	Description	Auth Required
POST	/api/auth/login	Login and get JWT token	No
GET	/api/auth/me	Get current user info	Yes (Bearer token)

Login Request:

```
{
  "username": "admin",
  "password": "admin123"
}
```

Login Response:

```
{
  "access_token": "eyJ3bG91ZDZlIHN1Iiw...\"",
  "token_type": "bearer"
}
```

WhatsApp / Messaging

Method	Path	Description	Auth Required
POST	/webhook/whatsapp	Twilio webhook endpoint	No
GET	/api/twilio/status	Check Twilio configuration	No
POST	/api/twilio/test	Send test WhatsApp message	No

Webhook Payload (from Twilio):

```
Body=@Raj fix server bug high priority
From=whatsapp:+919876543211
```

Webhook Response:

```
{
  "status": "ok",
  "intent": "TASK_ASSIGN",
  "language": "english"
}
```

Employees

Method	Path	Description	Auth Required
GET	/api/employees/	List all employees	No
GET	/api/employees/count	Employee count	No
POST	/api/employees/import	Import employees from CSV	No

Tasks

Method	Path	Description	Auth Required
GET	/api/tasks/	List all tasks	No
GET	/api/tasks/count	Task count	No
GET	/api/tasks/employee/{id}	Tasks for specific employee	No

Knowledge Base

Method	Path	Description	Auth Required
POST	/api/kb/upload	Upload PDF/TXT document	No
POST	/api/kb/search	Search KB with query	No

Internal (Admin Operations)

Method	Path	Description	Auth Required
POST	/internal/check-due-tasks	Check and remind overdue tasks	No
POST	/internal/check-periodic-	Check periodic follow-ups	No

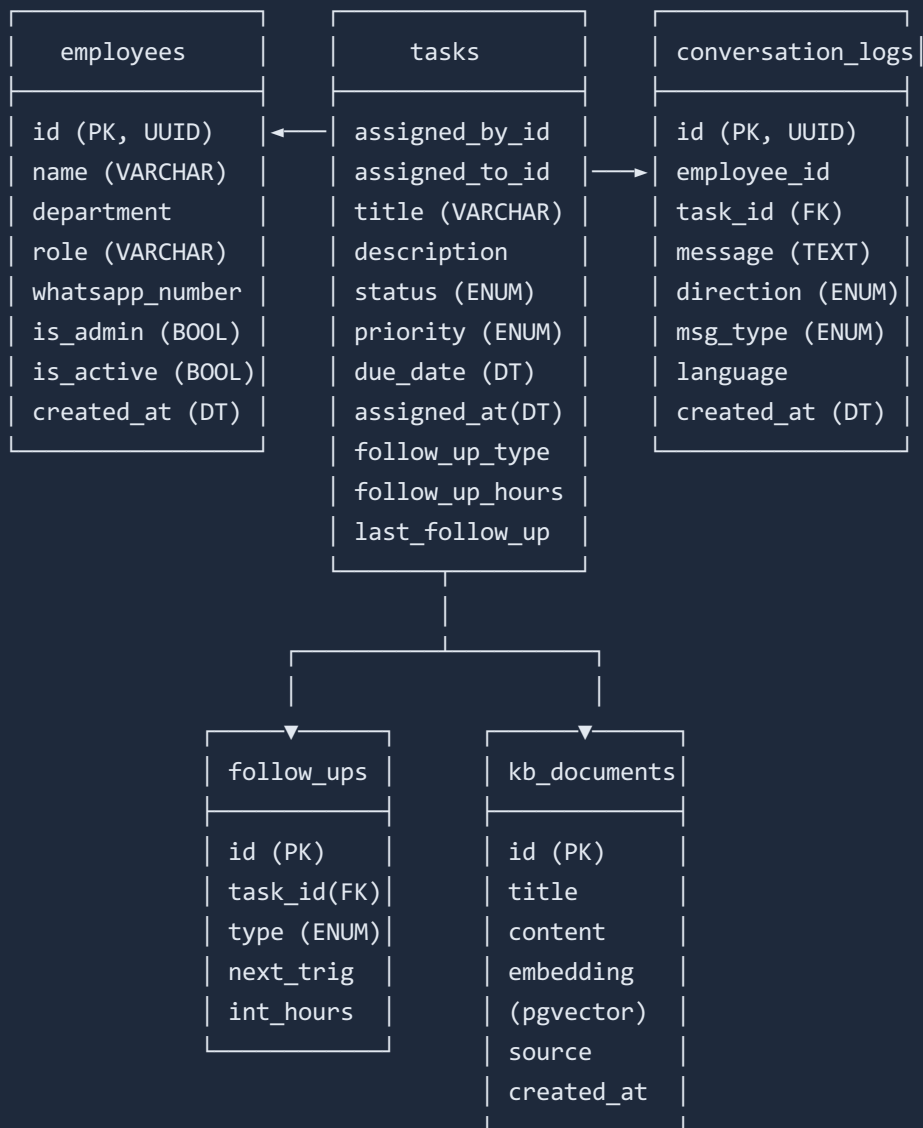
followups			
GET	/internal/stats	Get task statistics	No

Health

Method	Path	Description	Auth Required
GET	/health	System health check	No

9. Database Schema

Tables



Enum Values

Enum	Values
TaskStatus	pending , in_progress , done
Priority	low , medium , high
FollowUpType	none , periodic , deadline , escalation
Direction	inbound , outbound
MessageType	assignment , reply , trouble , system , follow_up
EscalationStatus	open , acknowledged , resolved

10. Production Deployment

10.1 Using Docker Compose

```
# 1. Clone and configure
cp .env.example .env
# Edit .env with production values

# 2. Start all services
docker-compose up -d --build

# 3. Verify
docker-compose ps
curl http://localhost:8000/health
curl http://localhost:80    # Frontend
```

10.2 Services

Service	Port	Purpose
db	5432	PostgreSQL 16 + pgvector
redis	6379	Caching & rate limiting
backend	8000	FastAPI API server
frontend	80	Nginx serving React SPA

10.3 Reverse Proxy (HTTPS)

Add nginx/Caddy in front of port 80:

```

server {
    listen 443 ssl;
    server_name yourdomain.com;

    ssl_certificate      /etc/ssl/certs/cert.pem;
    ssl_certificate_key  /etc/ssl/private/key.pem;

    location / {
        proxy_pass http://localhost:80;
    }

    location /webhook/ {
        proxy_pass http://localhost:8000;
        proxy_set_header Host $host;
    }
}

```

10.4 Environment for Production

```

# Twilio
TWILIO_ACCOUNT_SID=AC...
TWILIO_AUTH_TOKEN=your_real_token
TWILIO_WHATSAPP_NUMBER=+14155238886

# Database
DATABASE_URL=postgresql+psycpg2://wabot:STRONG_PASSWORD@db:5432/wabot
REDIS_URL=redis://redis:6379/0

# Auth
ADMIN_USERNAME=admin
ADMIN_PASSWORD=VERY_STRONG_PASSWORD_HERE
SECRET_KEY=$(openssl rand -hex 32)

# LLM (optional)
LLM_API_KEY=your_key
LLM_PROVIDER=anthropic

```

10.5 Database Migration

```

# Run Alembic migrations
docker-compose exec backend alembic upgrade head

```

11. Environment Variables

Variable	Required	Default
TWILIO_ACCOUNT_SID	No (for dev)	""
TWILIO_AUTH_TOKEN	No (for dev)	""
TWILIO_WHATSAPP_NUMBER	No (for dev)	""
LLM_API_KEY	No	""
LLM_PROVIDER	No	anthropic
LLM_MODEL	No	claude-3-5-haiku-latest
DATABASE_URL	Yes	postgresql+psycopg2://wabot:123@localhost:5433/wab
REDIS_URL	No	redis://localhost:6379/0
ADMIN_USERNAME	Yes	admin
ADMIN_PASSWORD	Yes	admin123
SECRET_KEY	Yes	change-me...
HOST	No	0.0.0.0

PORT

No

8000

12. Troubleshooting

Backend won't start

```
# Check Python version (3.12+ required)
python --version

# Reinstall dependencies
pip install --upgrade -r requirements.txt

# Test imports
python -c "from app.main import app; print('OK')"
```

Database connection refused

```
# Check container is running
docker ps | grep hrms-db

# Test connection
docker exec hrms-db psql -U wabot -d wabot -c "SELECT 1"

# Verify port
docker port hrms-db
```

pgvector not available

```
# Install pgvector in existing container
docker exec -it hrms-db /bin/bash
apt-get update && apt-get install -y build-essential postgresql-server-dev-16 git
cd /tmp
git clone --branch v0.8.0 https://github.com/pgvector/pgvector.git
cd pgvector
make && make install
psql -U wabot -d wabot -c "CREATE EXTENSION vector;"
```

Webhook not receiving messages

```
# 1. Check ngrok is running
ngrok http 8000

# 2. Verify webhook URL in Twilio console
# Must be: https://YOUR_NGROK_URL/webhook/whatsapp

# 3. Test endpoint directly
curl -X POST http://localhost:8000/webhook/whatsapp \
-d "Body=help&From=whatsapp:+919876543210"
```

Frontend not connecting to backend

```
# Verify vite proxy in vite.config.ts
# Must include: '/api', '/webhook', '/internal', '/health'

# Check backend is accessible
curl http://localhost:8000/health
```

Auth token expired

- Default expiry: 24 hours
- Change in `app/routers/auth.py` : `timedelta(hours=24)` → desired duration
- User is auto-redirected to login on 401

Twilio test fails with 400

- Ensure all 3 env vars are set: `TWILIO_ACCOUNT_SID` , `TWILIO_AUTH_TOKEN` , `TWILIO_WHATSAPP_NUMBER`
- Verify account SID starts with `AC`
- Check sandbox is joined on your phone

Quick Reference Card

Admin WhatsApp Commands

Message	Result
@Name task description	Assign task to employee

@Name task by Friday high	Assign with due date + priority
all tasks	See all team pending tasks
team pending	See all team pending tasks
follow up	Check and remind overdue tasks
help	Show admin command list

Employee WhatsApp Commands

Message	Result
done / ho gaya	Mark latest task complete
my tasks / pending	See your pending tasks
stuck / error / help	Get KB answer or escalate
help	Show employee command list

File Structure

```

wa-bot/
├── .env.example           # Environment template
├── docker-compose.yml    # Production deployment
├── SETUP.md              # Quick setup guide
├── PROJECT_DOC.md        # This document
├── sample_employees.csv   # Sample employee data
├── backend/
│   ├── Dockerfile
│   ├── requirements.txt
│   └── app/
│       ├── config.py      # Settings + env vars
│       ├── database.py    # SQLAlchemy setup
│       ├── main.py        # FastAPI app + routers
│       ├── models/        # SQLAlchemy models
│       │   ├── employee.py
│       │   ├── task.py
│       │   ├── conversation.py
│       │   ├── kb_document.py
│       │   └── escalation.py
│       ├── routers/       # API endpoints
│       │   ├── auth.py    # JWT login
│       │   └── twilio.py  # Twilio status/test

```

```
| | | | | webhook.py          # WhatsApp handler
| | | | | employees.py
| | | | | tasks.py
| | | | | kb.py
| | | | | └ internal.py
| | | | |
| | | | | └ services/          # Business logic
| | | | | | └ nlu.py          # NLP + keyword parser
| | | | | | └ whatsapp.py     # Twilio sender
| | | | | | └ task_manager.py
| | | | | | └ employee_svc.py
| | | | | | └ knowledge_base.py # RAG + pgvector
| | | | | └ utils/
| | | | | | └ helpers.py      # Mention/priority/date extractors
| | | └ tests/
| └ frontend/
|   └ Dockerfile
|   └ nginx.conf
|   └ package.json
|   └ vite.config.ts
|   └ tsconfig.json
|   └ src/
|     └ main.tsx
|     └ App.tsx              # Routes + ProtectedRoute
|     └ api/
|       └ client.ts          # API client + auth
|     └ components/
|       └ Layout.tsx
|       └ Sidebar.tsx        # Nav + logout
|     └ pages/
|       └ Login.tsx          # Auth page
|       └ Dashboard.tsx      # Stats + auto-refresh
|       └ Employees.tsx      # Search + filter + import
|       └ Tasks.tsx          # Status/priority filter + search
|       └ KnowledgeBase.tsx  # Upload + search
```